Pilot experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Pilot experiments

Preserve pilot defects and revisions while keeping new primary evidence separate.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **pilot** experiments

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of predefining an analysis plan

Reference example, not the only acceptable wording: Teacher explanation: An analysis plan records outcomes, units, exclusions, comparisons and uncertainty methods before primary results are inspected.

Deviations remain visible. Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory.

Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of research ethics and consent

Reference example, not the only acceptable wording: Teacher explanation: Research ethics considers purpose, burden, voluntary participation and consequences.

Approval to teach does not automatically authorise a research data collection. Use fictional records for this course.

For the supplied classroom policy, a real study needs responsible review, plain-language purpose, opt-in consent and withdrawal handling before collection. Do not implement real recruitment in this exercise.

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of ablation studies

Reference example, not the only acceptable wording: Teacher explanation: Ablation removes a component to investigate its contribution under a controlled comparison.

Interactions can make contributions depend on other components. Compare full system, without retrieval and without validation on identical cases and scoring.

Hold other settings fixed; record changed failure categories, not just totals.

Pictured end goal

Explain **pilot** experiments using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Predefining an analysis plan

Research ethics and consent

Ablation studies

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. A **pilot** checks **feasibility**, measurement and failure handling before the primary study. **Pilot** results used for revision are development **evidence**.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **pilot** experiments

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

A **pilot** checks **feasibility**, measurement and failure handling before the primary study. **Pilot** results used for revision are development **evidence**.

Run a small fixture set to test data capture, timing boundaries, rubric and analysis code. Fix defects and freeze the revised plan before new primary cases.

Pilot item	Complete supplied exercise contract
Main pilot	Five cases reveal wrong units in duration logs
Known defect	Unit handling is wrong; actual values and conversion direction are not supplied
Repair record	Preserve original log, corrected interpretation, unit definition and method version
Primary comparison	Use newly reserved cases after the revised method is fixed
Separate picture	Two reviewers score four practice answers; agree on three, disagree on one
Picture data role	Practice answers used for revision are excluded from final evaluation
Unknown evidence	No verified reference labels, disagreement cause, actual timings or final-study results

Main evidence ledger and checked key

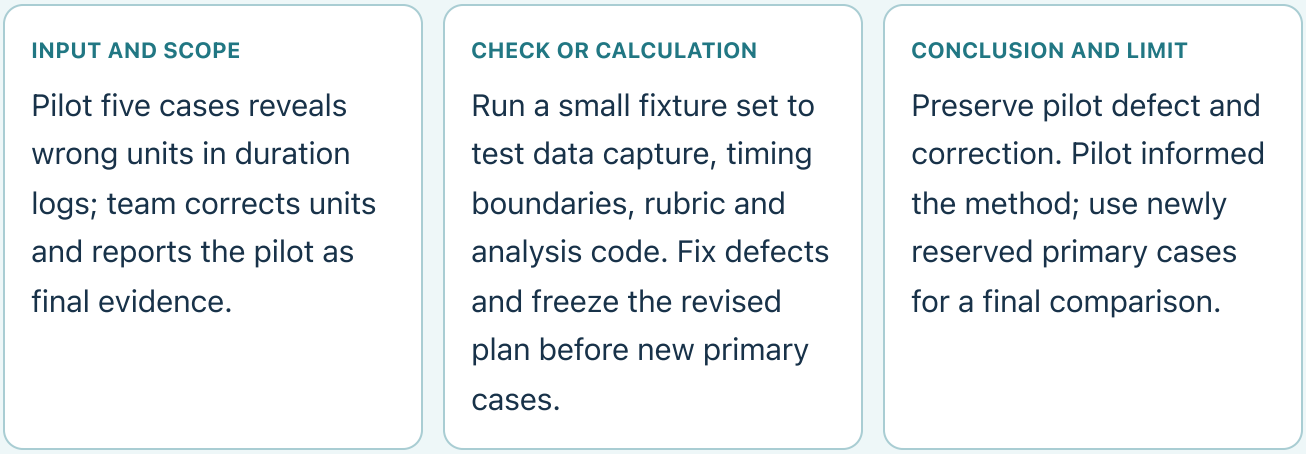
Main pilot step	Supported action	Unsupported claim
Five-case pilot exposes wrong duration units	Document defect and affected measurement	Pretend the original logs were correct
Correct unit handling	Record expected units, actual units and conversion evidence	Invent a numerical conversion without the missing values

Main pilot step	Supported action	Unsupported claim
Method revised using pilot	Retain pilot as development evidence	Call revised pilot outcomes an untouched final comparison
Freeze revised procedure	Record new version and planned measurement boundaries	Claim version label alone proves chronology
New primary cases	Reserve and evaluate under the fixed revised plan	Claim primary results before running the study

Main: preserve the five-case unit defect and correction history; values and unit direction are absent, so no numerical conversion is justified. Pilot-informed revision is development evidence. Freeze the revised method before newly reserved primary cases. Picture: three agreements and one disagreement among four practice answers gives 75% observed agreement, not correctness or long-run reliability.

The disagreement cause is unknown. Corrected practice answers remain excluded from final evaluation under this plan. Independent six-case pilot with two ambiguous labels calls for inspecting ambiguity, refining scoring instructions, checking the revised procedure and fixing the plan before primary data. Preserve difficult cases and revisions; do not claim a final model effect.

Pilot experiments — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Preserve **pilot** defect and correction. **Pilot** informed the method; use newly reserved primary cases for a final comparison.

Qualification: A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Pilot five cases reveals wrong units in duration logs; team corrects units and reports the **pilot** as final **evidence**.

Preserve **pilot** defect and correction. **Pilot** informed the method; use newly reserved primary cases for a final comparison.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Pilot audit	Checked interpretation
Pilot uncovers a defect	Useful evidence about procedure readiness
Units wrong but values absent	Do not invent a conversion or repaired timing
Pilot used to change method	Retain development role; do not call it untouched final evidence
Reviewers agree	May still share an incorrect interpretation
Reviewers disagree	Cause and correct label require inspection
Four-answer agreement is 75%	Observed agreement only, not proven long-run reliability
New primary study planned	No results exist until it is actually performed

Pilot audit	Checked interpretation
Ambiguous cases found	Revise and check rubric; do not silently delete difficult primary cases

Separate pictured facts and full reasoning

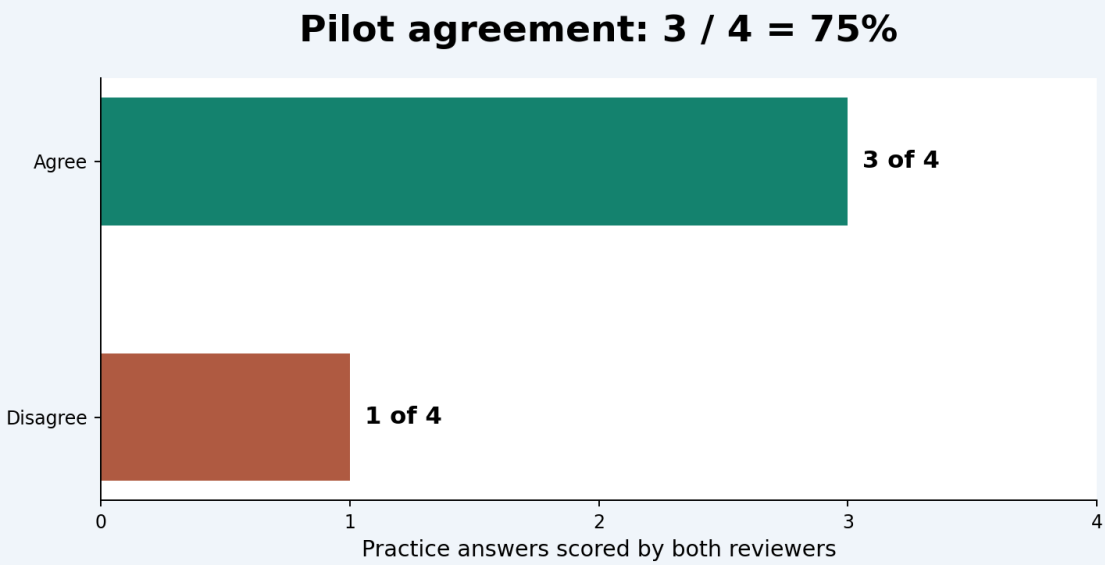
A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study.

Two reviewers score four practice answers. They agree on three and disagree on one; these answers are excluded from final evaluation.

The separate picture gives four practice answers scored by two reviewers. Three agreements out of four are 75% observed agreement; one disagreement is 25%. We are not told which answer differs, the label categories, who is right or why they disagreed. Inspect the actual scoring record and rubric before choosing a repair. Agreement does not establish correctness, and four answers do not establish long-run reliability.

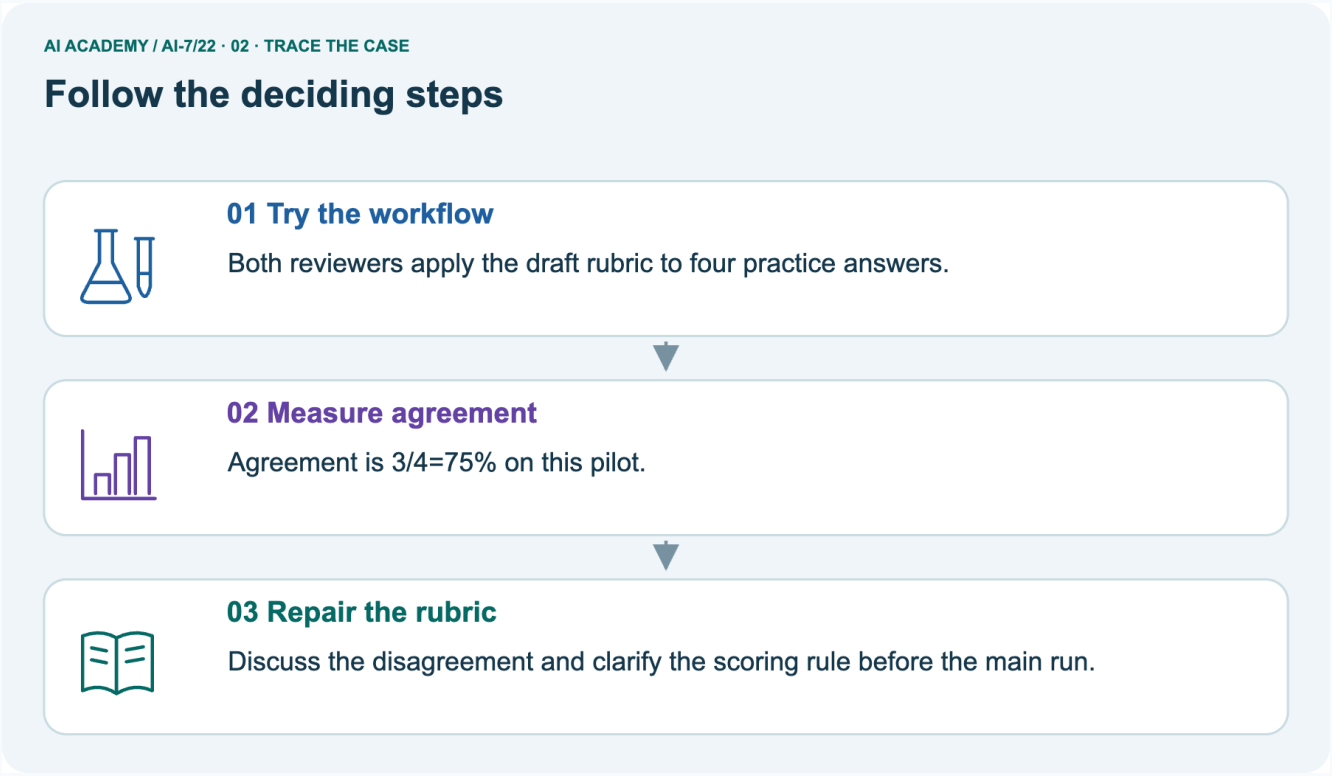
If these answers help revise the rubric or procedure, retain them as development evidence and keep them separate from untouched final evaluation. Adding the corrected practice cases to the final set would blur that role boundary.

Picture quantity	Supported value or limit
Answers scored by both reviewers	Four
Agree	Three; 3/4 = 75%
Disagree	One; 1/4 = 25%
Who is correct	Not supplied; agreement is not a reference label
Cause of disagreement	Unknown; inspect records and rubric
After used for revision	Development evidence, excluded from final evaluation under this plan
Long-run reliability / final effect	Not established by four practice answers



One disagreement can reveal a rubric problem; its cause is not supplied.
These four revised practice answers are development evidence, not untouched final cases.
Agreement is not correctness or a long-run reliability estimate.

Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Try the workflow	Both reviewers apply the draft rubric to four practice answers.
2. Measure agreement	Agreement is 3/4=75% on this pilot.
3. Repair the rubric	Discuss the disagreement and clarify the scoring rule before the main run.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **pilot** can reveal missing instructions, practical delays or ambiguous scoring. It is preparation, not an untouched final estimate after the team uses it to revise methods.

Four **pilot** answers do not establish long-run reviewer reliability or a stable performance effect.

Change: Add the corrected **pilot** answers into final evaluation.

Prediction: Declare them previously exposed rather than untouched final **evidence**.

Reason: Rubric development already used those answers.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Pilot	A preliminary study of the procedure
Feasibility	Whether the planned method can be carried out
Measurement defect	A failure in collecting the intended outcome

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

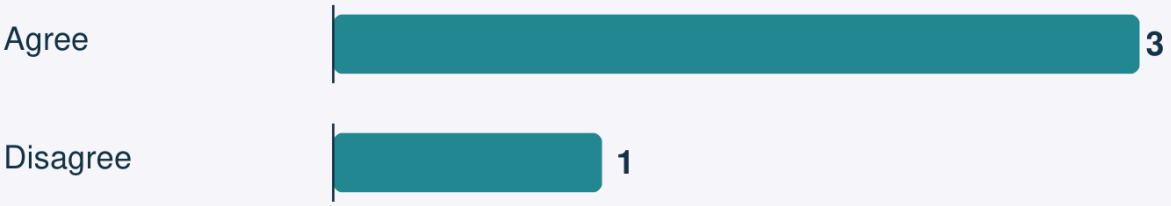
Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Pilot experiments

The facts we will use

AI ACADEMY / AI-7/22

Reviewer agreement on four pilot answers



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/22

Pilot experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Try the workflow

Both reviewers apply the draft rubric to four practice answers.



2. Measure agreement

Agreement is 3/4=75% on this pilot.



3. Repair the rubric

Discuss the disagreement and clarify the scoring rule before the main run.

CHANGE ONE THING

Add the corrected pilot answers into final evaluation.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: **Pilot** five cases reveals wrong units in duration logs; team corrects units and reports the **pilot** as final **evidence**. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Preserve **pilot** defect and correction. **Pilot** informed the method; use newly reserved primary cases for a final comparison.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Pilot** experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate reviewer agreement on the four **pilot** answers. What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?

Both reviewers apply the draft rubric to four practice answers. Agreement is 3/4=75% on this **pilot**. Discuss the disagreement and clarify the scoring rule before the main run.

Changed case: Declare them previously exposed rather than untouched final **evidence**. Rubric development already used those answers.

Changed-case reasoning: Declare them previously exposed rather than untouched final **evidence**. Rubric development already used those answers.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main: preserve the five-case unit defect and correction history; values and unit direction are absent, so no numerical conversion is justified. Pilot-informed revision is development evidence. Freeze the revised method before newly reserved primary cases. Picture: three agreements and one disagreement among four practice answers gives 75% observed agreement, not correctness or long-run reliability.

The disagreement cause is unknown. Corrected practice answers remain excluded from final evaluation under this plan. Independent six-case pilot with two ambiguous labels calls for inspecting ambiguity, refining scoring instructions, checking the revised procedure and fixing the plan before primary data. Preserve difficult cases and revisions; do not claim a final model effect.

Diagnose the current misconceptions

The worked result establishes every future case: A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: **Pilot** six cases exposes ambiguous labels in two. What should happen before the primary study?

Teacher explanation: Clarify label rubric, reassess **pilot** as development, record versions and reserve fresh primary **evidence** under the frozen rubric.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A **pilot** checks **feasibility**, measurement and failure handling before the primary study. **Pilot** results used for revision are development **evidence**.

Run a small fixture set to test data capture, timing boundaries, rubric and analysis code. Fix defects and freeze the revised plan before new primary cases.

W2 Trace the worked case

Pilot five cases reveals wrong units in duration logs; team corrects units and reports the **pilot** as final **evidence**. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Preserve **pilot** defect and correction. **Pilot** informed the method; use newly reserved primary cases for a final comparison.

W3 Audit the limit

A report omits this qualification: A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Preserve **pilot** defect and correction. **Pilot** informed the method; use newly reserved primary cases for a final comparison.

Qualification: A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Pilot six cases exposes ambiguous labels in two. What should happen before the primary study?

Teacher explanation: Clarify label rubric, reassess **pilot** as development, record versions and reserve fresh primary **evidence** under the frozen rubric.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Clarify label rubric, reassess **pilot** as development, record versions and reserve fresh primary **evidence** under the frozen rubric. Require intermediate reasoning and the stated scope.

Fresh retry: **Pilot** reveals skipped failures not logged. Expected reasoning: Repair capture and missing-outcome handling; verify with fixtures before collecting primary data.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled **pilot** experiments **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **pilot** experiments.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Power and resource tradeoffs. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/23, ai-7/25, ai-7/28. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **pilot** experiments

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: **Pilot** five cases reveals wrong units in duration logs; team corrects units and reports the **pilot** as final **evidence**.

Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Preserve **pilot** defect and correction. **Pilot** informed the method; use newly reserved primary cases for a final comparison.

Check every learner before discussion: Use the worked case for **Pilot** experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add the corrected **pilot** answers into final evaluation. Require a **prediction** and a reason before comparing. Expected result: Declare them previously exposed rather than untouched final **evidence**.

Reason: Rubric development already used those answers.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

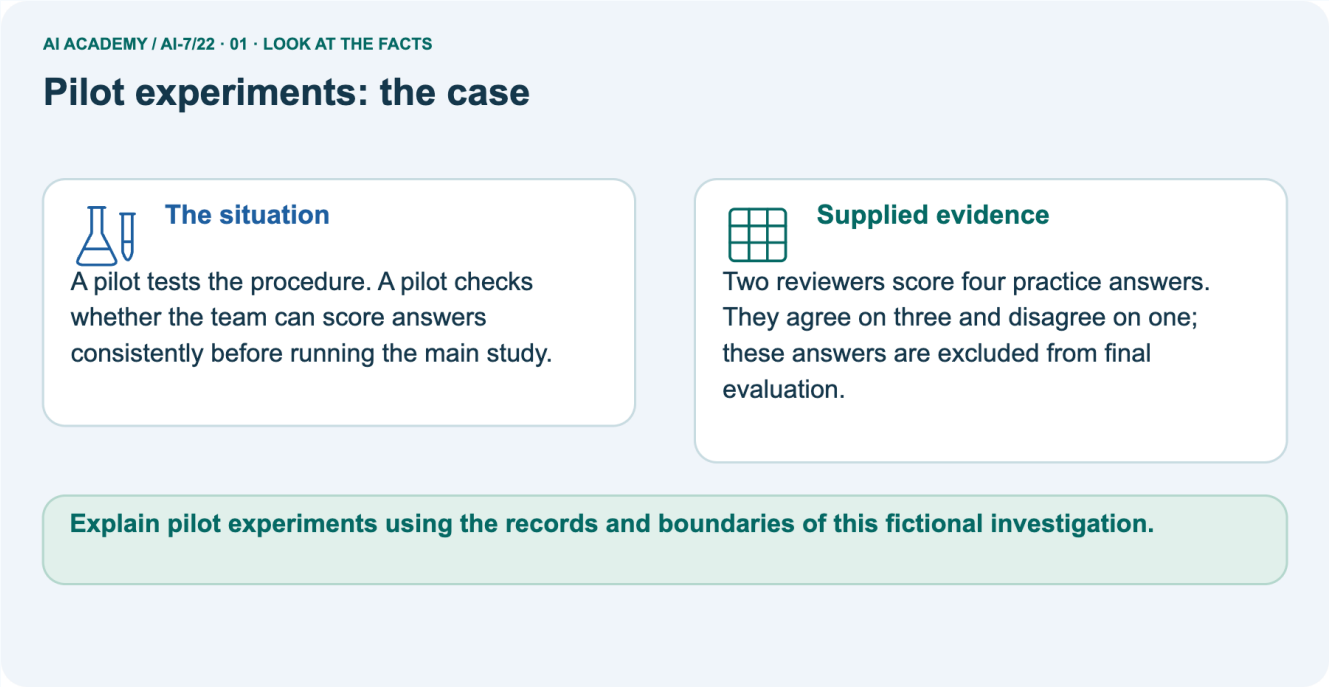
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Pilot experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

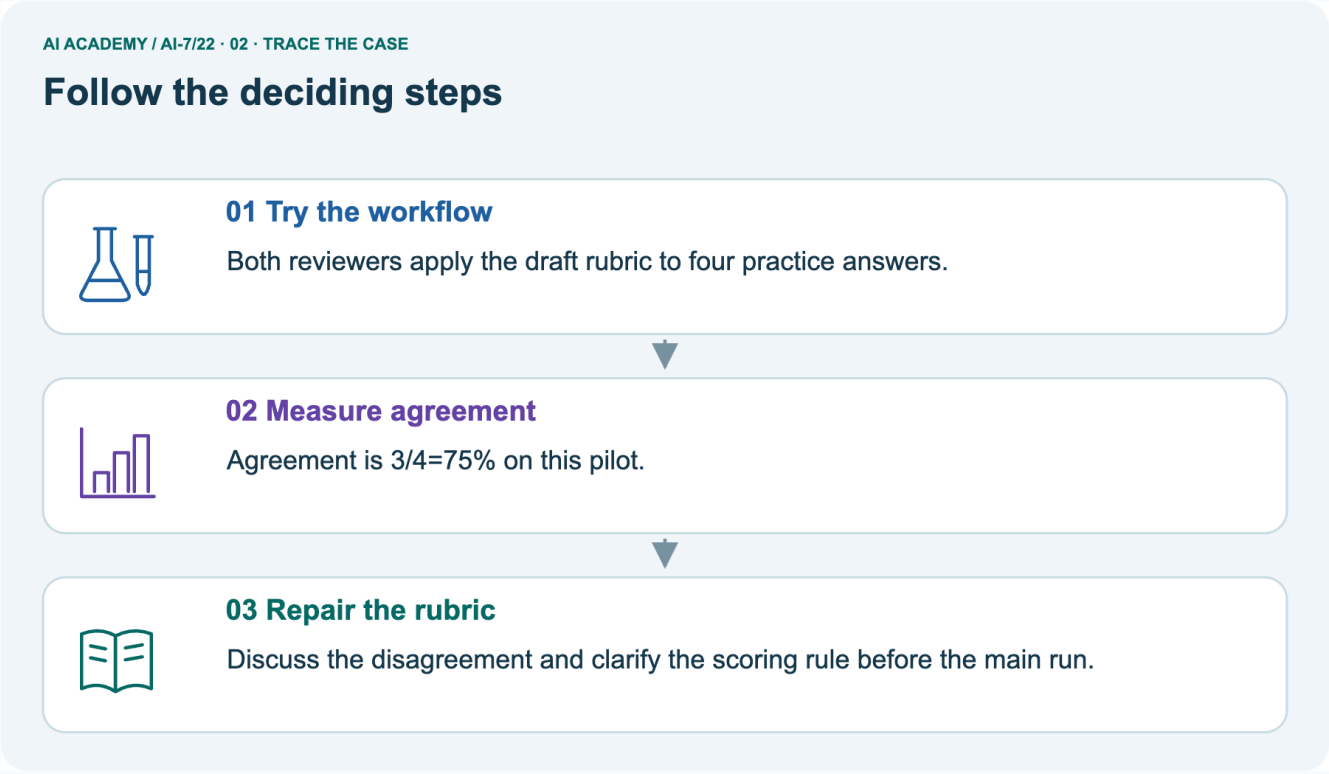
A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study. Supplied facts: Two reviewers score four practice answers.

They agree on three and disagree on one; these answers are excluded from final evaluation.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



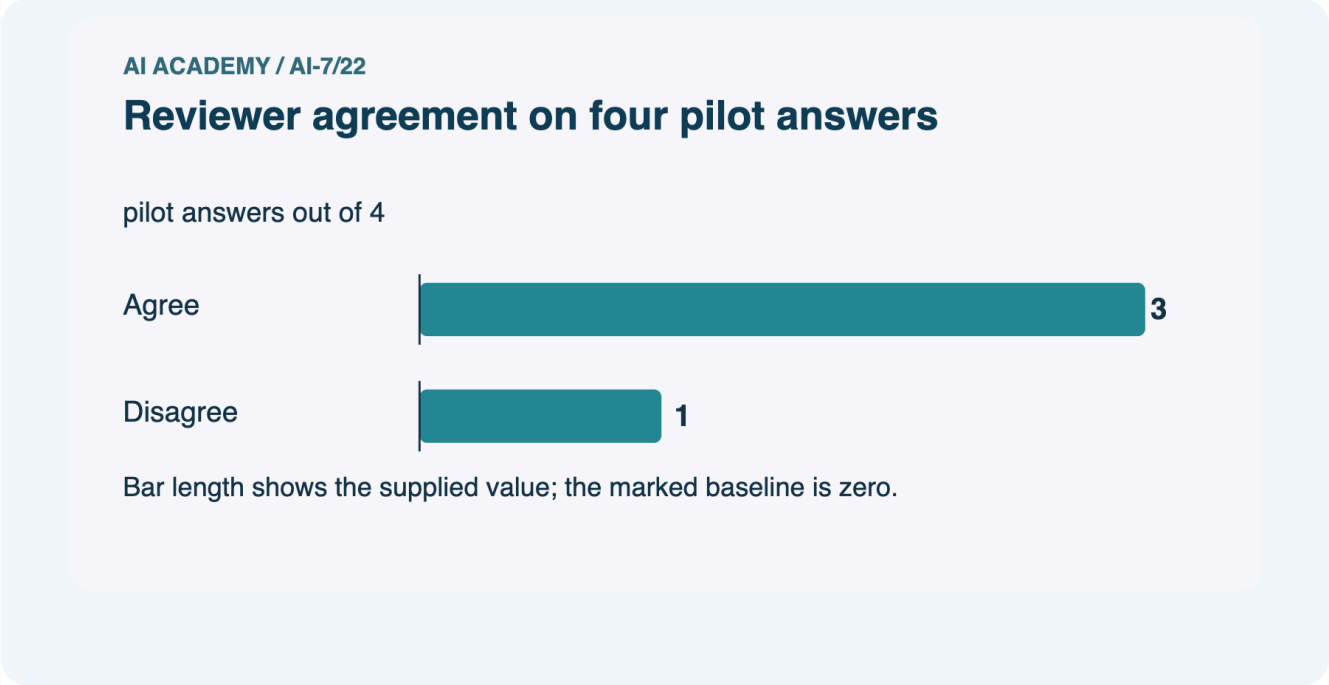
Follow the labelled picture with the matching explanation. Open the vector diagram

Try the workflow: Both reviewers apply the draft rubric to four practice answers. → Measure agreement: Agreement is 3/4=75% on this **pilot**. → Repair the rubric: Discuss the disagreement and clarify the scoring rule before the main run.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Reviewer agreement on four pilot answers



Follow the labelled picture with the matching explanation. Open the vector diagram

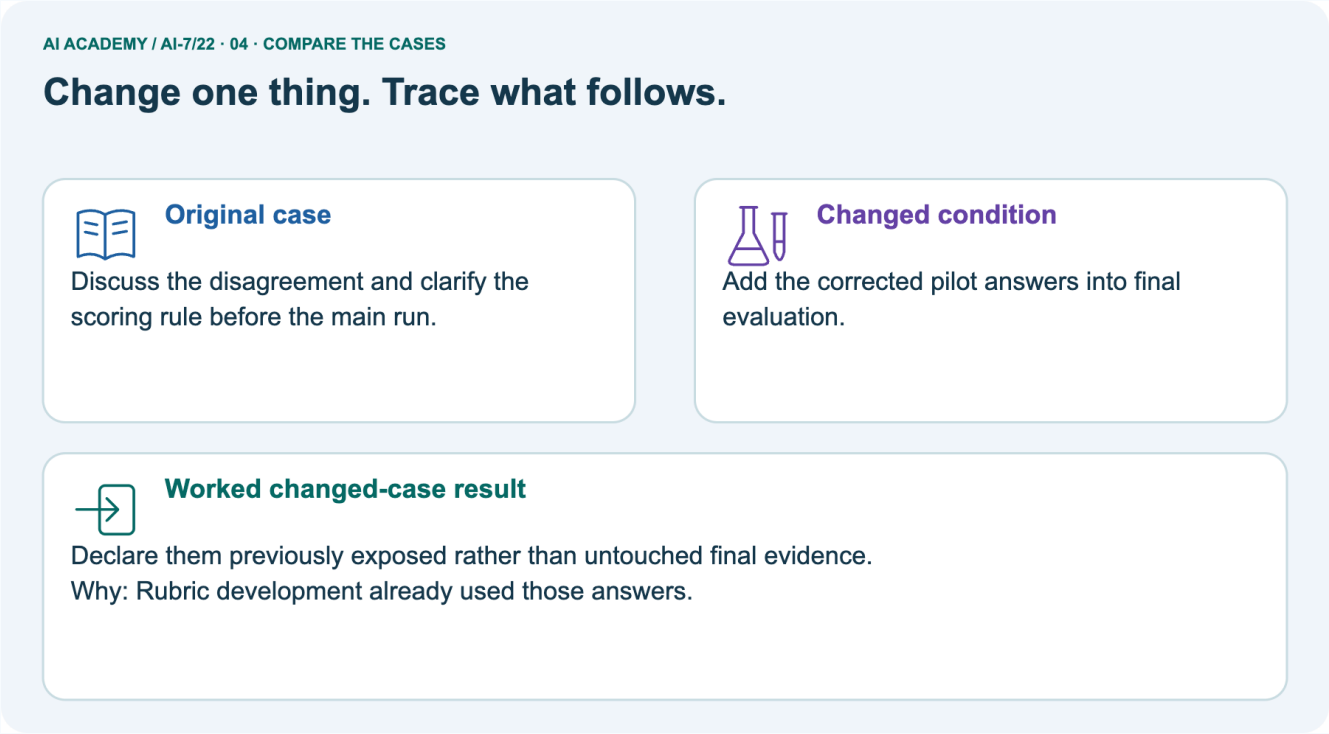
Reviewer agreement on four **pilot** answers. Units: **pilot** answers out of 4. Agree: 3. Disagree: 1.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Discuss the disagreement and clarify the scoring rule before the main run.
Changed condition: Add the corrected **pilot** answers into final evaluation.

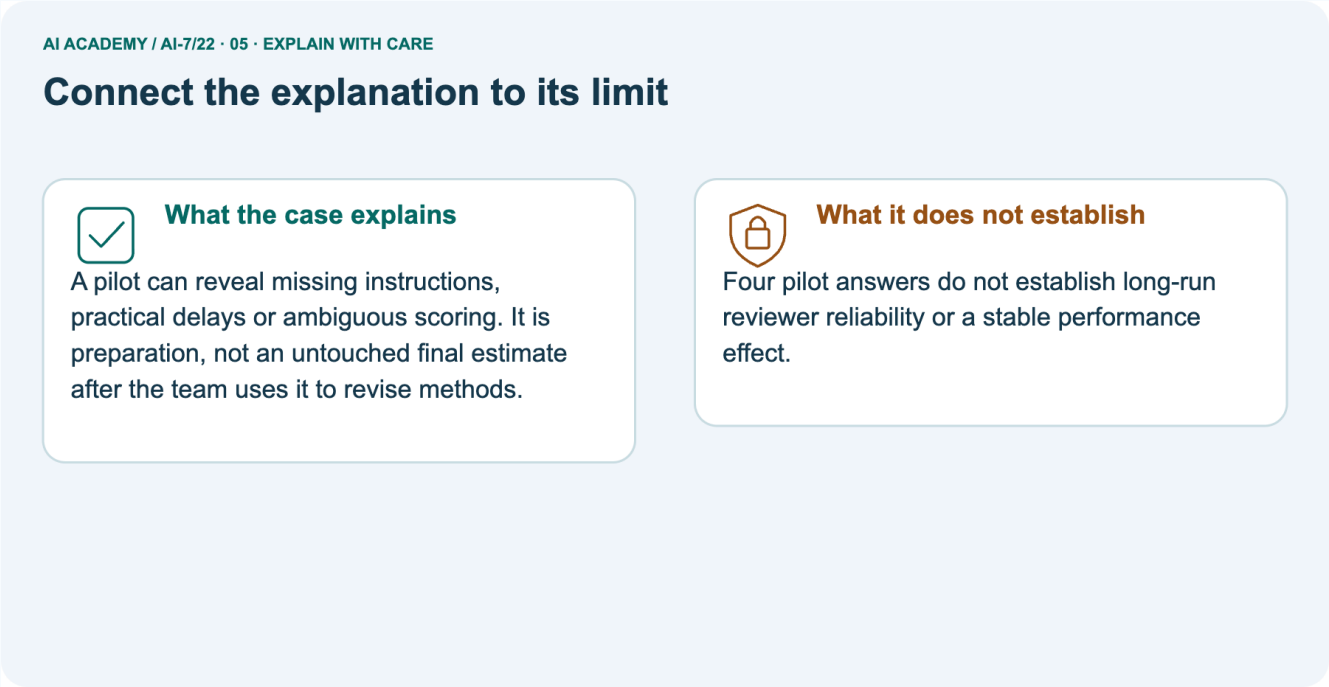
Expected result: Declare them previously exposed rather than untouched final **evidence**.
Reason: Rubric development already used those answers.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Rubric development already used those answers.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **pilot** can reveal missing instructions, practical delays or ambiguous scoring. It is preparation, not an untouched final estimate after the team uses it to revise methods.

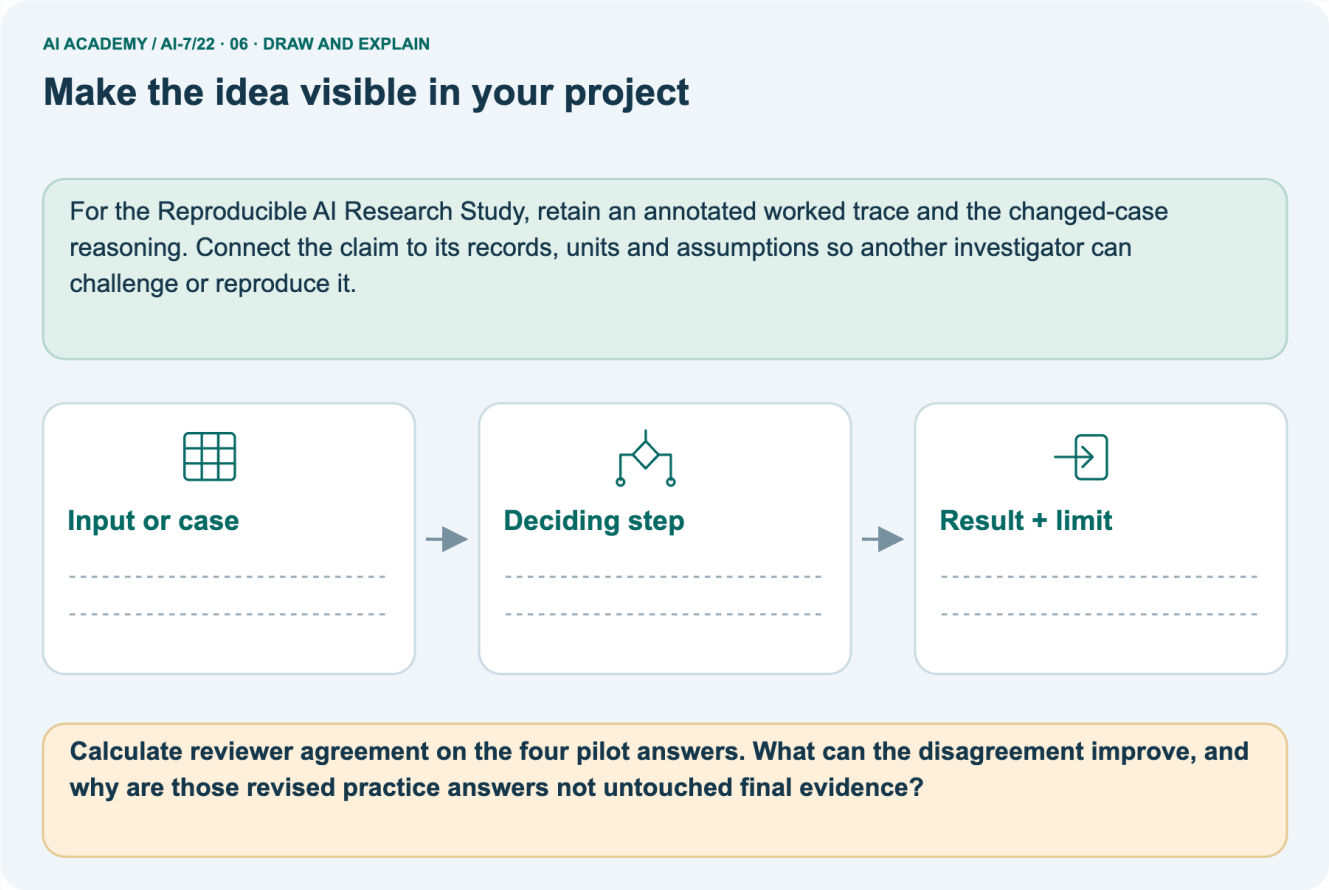
Limit: Four **pilot** answers do not establish long-run reviewer reliability or a stable performance effect.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Four **pilot** answers do not establish long-run reviewer reliability or a stable performance effect.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate reviewer agreement on the four **pilot** answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Both reviewers apply the draft rubric to four practice answers. Agreement is 3/4=75% on this **pilot**. Discuss the disagreement and clarify the scoring rule before the main run.

Changed case: Declare them previously exposed rather than untouched final **evidence**. Rubric development already used those answers.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Pilot experiments: the case

02. Follow the deciding steps

03. Reviewer agreement on four pilot answers

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.